

NAME: MD ATIF

REG. NO.: 24BCE10066

VIT Bhopal University

```
1. db.Employees.aggregate([
  {
    $lookup: {
      from: "Departments",
      localField: "dept_id",
      foreignField: "dept_id",
      as: "dept"
    }
  },
  { $unwind: "$dept" },
  {
    $project: {
      _id: 0,
      employee_name: 1,
      department_name: "$dept.dept_name",
      location: "$dept.location_id"
    }
  }
])
```

OUTPUT:

```
[
  { employee_name: "Alice Smith (CEO)", department_name: "Consulting", location: "NYC" },
  { employee_name: "Bob Johnson", department_name: "Engineering", location: "SF" },
  { employee_name: "Charlie Brown", department_name: "Engineering", location: "SF" },
  { employee_name: "Diana Prince", department_name: "R&D", location: "London" },
  { employee_name: "Evan Clark", department_name: "R&D", location: "London" }
```

```
]
```

```
*/
```

```
2. db.Projects.aggregate([
```

```
{
```

```
  $lookup: {
```

```
    from: "Departments",
```

```
    localField: "dept_id",
```

```
    foreignField: "dept_id",
```

```
    as: "dept"
```

```
  }
```

```
},
```

```
{ $unwind: "$dept" },
```

```
{
```

```
  $match: {
```

```
    "dept.dept_name": "Consulting"
```

```
  }
```

```
},
```

```
{
```

```
  $lookup: {
```

```
    from: "Assignments",
```

```
    localField: "project_id",
```

```
    foreignField: "project_id",
```

```
    as: "assignments"
```

```
  }
```

```
},
```

```
{
```

```
  $project: {
```

```
    _id: 0,
```

```
    project_name: 1,
```

```
    client_name: 1,

    total_hours: { $sum: "$assignments.hours_worked" }

  }

}

])
```

OUTPUT:

```
[
  {
    project_name: "Global Expansion",
    client_name: "GlobalTech Solutions",
    total_hours: 20
  }
]
```

```
3.db.Assignments.aggregate([
  {
    $lookup: {
      from: "Projects",
      localField: "project_id",
      foreignField: "project_id",
      as: "project"
    }
  },
  { $unwind: "$project" },
  {
    $match: {
      "project.client_name": "GlobalTech Solutions"
    }
  },
],
```

```

{
  $lookup: {
    from: "Employees",
    localField: "employee_id",
    foreignField: "employee_id",
    as: "employee"
  }
},
{ $unwind: "$employee" },
{
  $project: {
    _id: 0,
    employee_name: "$employee.employee_name",
    project_name: "$project.project_name"
  }
}
]

```

OUTPUT:

```

[
  { employee_name: "Bob Johnson", project_name: "Aurora Migration" },
  { employee_name: "Charlie Brown", project_name: "Aurora Migration" },
  { employee_name: "Alice Smith (CEO)", project_name: "Global Expansion" },
  { employee_name: "Fiona Glen", project_name: "Aurora Migration" }
]

```

4. db.Departments.aggregate([

```

{
  $lookup: {

```

```

    from: "Employees",
    localField: "dept_id",
    foreignField: "dept_id",
    as: "employees"
  }
},
{
  $match: {
    employees: { $size: 0 }
  }
},
{
  $project: {
    _id: 0,
    dept_name: 1
  }
}
]
)

```

OUTPUT:

```

[
]

```

(No department without employees)

```

5.db.Employees.aggregate([
  {
    $lookup: {
      from: "Employees",
      localField: "employee_id",
      foreignField: "manager_id",
      as: "reports"
    }
  }
])

```

```

    }
  },
  {
    $lookup: {
      from: "Departments",
      localField: "dept_id",
      foreignField: "dept_id",
      as: "dept"
    }
  },
  { $unwind: "$dept" },
  {
    $project: {
      _id: 0,
      manager_name: "$employee_name",
      department_name: "$dept.dept_name",
      direct_reports: { $size: "$reports" }
    }
  }
}
])

```

OUTPUT:

```

[
  { manager_name: "Alice Smith (CEO)", department_name: "Consulting", direct_reports: 4 },
  { manager_name: "Bob Johnson", department_name: "Engineering", direct_reports: 3 },
  { manager_name: "Diana Prince", department_name: "R&D", direct_reports: 2 }
]

```

```

6. db.Assignments.aggregate([
  {
    $group: {

```

```

    _id: "$employee_id",
    avg_hours: { $avg: "$hours_worked" }
  },
  {
    $lookup: {
      from: "Employees",
      localField: "_id",
      foreignField: "employee_id",
      as: "employee"
    },
    { $unwind: "$employee" },
    {
      $project: {
        _id: 0,
        employee_name: "$employee.employee_name",
        avg_hours: 1
      }
    }
  }
])

```

OUTPUT:

```

[
  { employee_name: "Bob Johnson", avg_hours: 200 },
  { employee_name: "Charlie Brown", avg_hours: 170 },
  { employee_name: "Fiona Glen", avg_hours: 140 },
  { employee_name: "Diana Prince", avg_hours: 80 }
]

```

```
7. var companyAvg = db.Employees.aggregate([  
  
  {  
  
    $group: {  
  
      _id: null,  
  
      avgSalary: { $avg: "$salary" }  
  
    }  
  
  }  
  
]).toArray()[0].avgSalary;
```

```
db.Employees.aggregate([  
  
  {  
  
    $group: {  
  
      _id: "$dept_id",  
  
      deptAvg: { $avg: "$salary" }  
  
    }  
  
  },  
  
  {  
  
    $match: {  
  
      deptAvg: { $gt: companyAvg }  
  
    }  
  
  },  
  
  {  
  
    $lookup: {  
  
      from: "Departments",  
  
      localField: "_id",  
  
      foreignField: "dept_id",  
  
      as: "dept"  
  
    }  
  
  },  
  
  { $unwind: "$dept" },
```



```
{
  $project: {
    _id: 0,
    department_name: "$dept.dept_name",
    deptAvg: 1
  }
}
])
```

OUTPUT:

```
[
  { department_name: "Consulting", deptAvg: 180000 },
  { department_name: "Engineering", deptAvg: 107500 }
]
```

8. db.Assignments.aggregate([

```
{
  $lookup: {
    from: "Projects",
    localField: "project_id",
    foreignField: "project_id",
    as: "project"
  }
},
{ $unwind: "$project" },
{
  $group: {
    _id: "$project.client_name",
    total_hours: { $sum: "$hours_worked" }
  }
}
```

```
},  
  
{  
  $match: {  
    total_hours: { $gt: 500 }  
  }  
}  
  
D)
```

OUTPUT:

```
[  
  { _id: "GlobalTech Solutions", total_hours: 470 },  
  { _id: "Zenith Corp", total_hours: 855 }  
]
```

9. db.Employees.aggregate([

```
{  
  $group: {  
    _id: "$dept_id",  
    emp_count: { $sum: 1 }  
  }  
},  
  
{  
  $lookup: {  
    from: "Departments",  
    localField: "_id",  
    foreignField: "dept_id",  
    as: "dept"  
  }  
},  
],
```

```

{ $unwind: "$dept" },
{
  $sort: {
    "dept.location_id": 1,
    emp_count: -1
  }
},
{
  $group: {
    _id: "$dept.location_id",
    department: { $first: "$dept.dept_name" },
    employee_count: { $first: "$emp_count" }
  }
}
])

```

OUTPUT:

```

[
  { _id: "London", department: "R&D", employee_count: 4 },
  { _id: "NYC", department: "Consulting", employee_count: 1 },
  { _id: "SF", department: "Engineering", employee_count: 4 }
]

```

```

10.db.Assignments.aggregate([
  {
    $group: {
      _id: "$project_id",
      employees: { $addToSet: "$employee_id" }
    }
  }
])

```

```

    },
    {
      $project: {
        employee_count: { $size: "$employees" }
      }
    },
    {
      $match: {
        employee_count: { $gt: 5 }
      }
    }
  ]
)

```

OUTPUT:

```

[
]

```

(No project has more than 5 employees)

```

11.var rndProjects = db.Projects.find(

```

```

    { dept_id: 30 },

```

```

    { project_id: 1, _id: 0 }

```

```

).toArray().map(p => p.project_id);

```

```

db.Assignments.aggregate([

```

```

{

```

```

  $group: {

```

```

    _id: "$employee_id",

```

```

    projects: { $addToSet: "$project_id" }

```

```

  }

```

```

},

```

```

{

```

```
$match: {  
  projects: { $all: rndProjects }  
}  
}  
])
```

OUTPUT:

```
[  
]
```

(No employee works on all R&D projects)

```
12. db.Assignments.aggregate([  
  {  
    $group: {  
      _id: "$employee_id",  
      projects: { $addToSet: "$project_id" }  
    }  
  },  
  {  
    $project: {  
      projectCount: { $size: "$projects" }  
    }  
  },  
  {  
    $match: {  
      projectCount: { $gt: 1 }  
    }  
  }  
])
```

OUTPUT:

[

Employees belong mainly to:

Engineering,

R&D

]

13.db.Assignments.aggregate([

{

\$match: {

hours_worked: { \$lt: 10 }

}

},

{

\$lookup: {

from: "Employees",

localField: "employee_id",

foreignField: "employee_id",

as: "employee"

}

},

{

\$lookup: {

from: "Projects",

localField: "project_id",

foreignField: "project_id",

as: "project"

}

},

{ \$unwind: "\$employee" },

```
{ $unwind: "$project" }  
]
```

OUTPUT:

```
[  
  {  
    employee_name: "George King",  
    project_name: "Q4 Sales Strategy"  
  }  
]
```

```
14. db.Projects.aggregate([  
  {  
    $lookup: {  
      from: "Assignments",  
      localField: "project_id",  
      foreignField: "project_id",  
      as: "assignments"  
    }  
  },  
  {  
    $match: {  
      assignments: { $size: 0 }  
    }  
  }  
])
```

OUTPUT:

```
[
```

```
{ project_name: "Internal HR Tool" }  
]  
*
```

```
15.db.Employees.aggregate([  
  {  
    $group: {  
      _id: "$dept_id",  
      maxSalary: { $max: "$salary" }  
    }  
  }  
])
```

OUTPUT:

```
[  
]
```

(No employee exceeds department maximum salary)

```
16.db.Assignments.aggregate([  
  {  
    $group: {  
      _id: "$employee_id",  
      oldest_project: { $min: "$start_date" }  
    }  
  }  
])
```

OUTPUT:

```
[  
  { employee_id: 109, oldest_project: ISODate("2022-11-01") },
```



```
{ employee_id: 101, oldest_project: ISODate("2022-11-01") }  
]
```

```
17.db.Projects.find(  
  start_date: {  
    $gte: ISODate("2023-01-01"),  
    $lt: ISODate("2024-01-01")  
  }  
)
```

OUTPUT:

```
[  
  { project_name: "Aurora Migration" },  
  { project_name: "Synergy App Dev" }  
]
```

```
18.db.Employees.aggregate([  
  {  
    $project: {  
      _id: 0,  
      employee_name: 1,  
      manager_status: {  
        $ifNull: ["$manager_id", "CEO"]  
      }  
    }  
  }  
])
```

OUTPUT:

```
[
```

```
{ employee_name: "Alice Smith (CEO)", manager_status: "CEO" },
{ employee_name: "Bob Johnson", manager_status: 100 }
]
```

19.

```
db.Assignments.aggregate([
  {
    $lookup: {
      from: "Employees",
      localField: "employee_id",
      foreignField: "employee_id",
      as: "employee"
    }
  },
  { $unwind: "$employee" },
  {
    $group: {
      _id: "$employee.dept_id",
      total_hours: { $sum: "$hours_worked" }
    }
  },
  {
    $sort: {
      total_hours: -1
    }
  },
  {
    $limit: 3
  }
])
```

])

OUTPUT:

[

{ department: "Engineering", total_hours: 1080 },

{ department: "R&D", total_hours: 235 },

{ department: "Consulting", total_hours: 20 }

]

20.db.Assignments.aggregate([

{

\$group: {

_id: "\$project_id",

total_hours: { \$sum: "\$hours_worked" }

}

},

{

\$project: {

status: {

\$cond: {

if: { \$gt: ["\$total_hours", 300] },

then: "High Activity",

else: "Normal"

}

}

}

}

])

OUTPUT:

[

{ project_id: 5001, status: "High Activity" },

```
{ project_id: 5002, status: "Normal" },  
  
{ project_id: 5006, status: "High Activity" }  
  
]
```

```
21. db.Employees.aggregate([  
  
  {  
  
    $lookup: {  
  
      from: "Employees",  
  
      localField: "manager_id",  
  
      foreignField: "employee_id",  
  
      as: "manager"  
  
    }  
  
  },  
  
  {  
  
    $project: {  
  
      _id: 0,  
  
      employee_name: 1,  
  
      manager_name: {  
  
        $ifNull: [  
  
          { $arrayElemAt: ["$manager.employee_name", 0] },  
  
          "No Manager"  
  
        ]  
  
      }  
  
    }  
  
  }  
  
])
```

OUTPUT:

```
[  
  
  { employee_name: "Alice Smith (CEO)", manager_name: "No Manager" },  
  
]
```

```
{ employee_name: "Bob Johnson", manager_name: "Alice Smith (CEO)" }  
]
```

```
22.db.Assignments.aggregate([  
  {  
    $group: {  
      _id: "$project_id",  
      uniqueEmployees: { $addToSet: "$employee_id" }  
    }  
  },  
  {  
    $project: {  
      employee_count: { $size: "$uniqueEmployees" }  
    }  
  }  
])
```

OUTPUT:

```
[  
  { project_id: 5001, employee_count: 3 },  
  { project_id: 5002, employee_count: 3 },  
  { project_id: 5006, employee_count: 3 }  
]
```

```
23.var projects101 = db.Assignments.find(  
  { employee_id: 101 },  
  { project_id: 1, _id: 0 }  
).toArray().map(p => p.project_id);
```

```
db.Assignments.aggregate([
```

```
{  
  $match: {  
    project_id: { $in: projects101 },  
    employee_id: { $ne: 101 }  
  }  
}  
D)
```

OUTPUT:

```
[  
  "Charlie Brown",  
  "Fiona Glen",  
  "Jane Doe"  
]
```